# Contrastive Learning

$x \xrightarrow{NN} emb(x)$

↑

image of person

NN goal:

if $x_1$ and $x_2$ - photos of the same person =>

$dist(emb(x_1), emb(x_2))$ - low

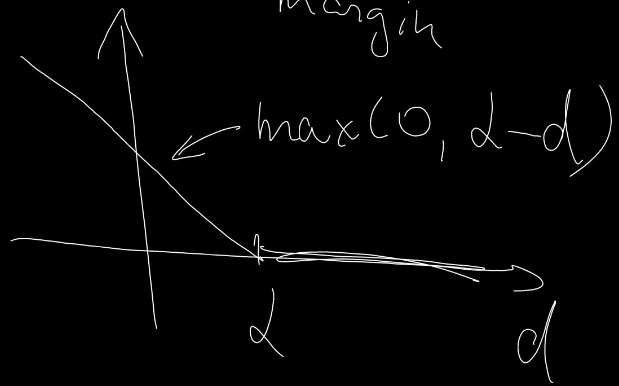
if $x_1$ and $x_2$ - different persons =>

$dist(emb(x_1), emb(x_2))$ - high

## Constrastive loss:

$$\mathcal{L}(x_1, x_2, y) = (1-y)\frac{1}{2}\,\mathrm{dist}(\mathrm{emb}(x_1), \mathrm{emb}(x_2)) + y\,\max(0, \underbrace{\alpha}_{\text{margin}} - \mathrm{dist}(\mathrm{emb}(x_1), \mathrm{emb}(x_2))) \rightarrow \text{min emb}$$

$$y = \begin{cases} 0, & \text{the same person} \\ 1, & \text{different persons} \end{cases}$$



$$dist(x,y) = \|x-y\|_2^2 = \|x\|^2 + \|y\|^2 - 2x^T y$$

$$\hat{x} = \frac{x}{\|x\|}, \quad \hat{y} = \frac{y}{\|y\|}$$

$$dist(x,y) = \|\hat{x}-\hat{y}\|_2^2 = 1 + 1 - 2\hat{x}^T\hat{y} =$$

$$= 2 - 2\underbrace{\cos\theta}_{\frac{x^T y}{\|x\|\cdot\|y\|}}$$

# Triplet Loss

a – anchor
p – positive
n – negative

$$\mathcal{L}(a,p,n) =$$
$$= \max(0,\ d(a,p) + \alpha - d(a,n)) \longrightarrow \min$$

We want: $d(a,p) + \alpha < d(a,n)$

PK sampler (hard examples mining)

in one mini-batch sample P persons and
K photos for each person

for each anchor (person) take farthest example as positive
closest example as negative

# ArcFace

Inference: do not use $w$,
use $\frac{emb(x)^T emb(y)}{\|emb(x)\| \cdot \|emb(y)\|}$

$w_i$ — class center for person $i$

$$\cos\theta_{ij} = \frac{w_i^T emb(x_j)}{\|w_i\| \cdot \|emb(x_j)\|}$$

$$\text{Loss} = -\log \frac{\exp(\tau \cdot \cos(\theta_{y(i)x_i} + m))}{\exp(\tau \cos(\theta_{y(i)x_i} + m)) + \sum_{y \neq y(i)} \exp(\tau \cos\theta_{y x_i})} \to \min_{w,\, emb}$$

# CLIP (Contrastive Language-Image Pre-training)

(img, txt) - image and its description

img_emb = VisionTransformer(img)

txt_emb = Transformer(txt)

$$\text{score}(i, j) = \frac{\text{img\_emb}_i^T \, \text{txt\_emb}_j}{\|\text{img\_emb}_i\| \cdot \|\text{txt\_emb}_j\|}$$

Sample $N$ pairs $(img_i, txt_i)$, $i = 1, \dots, N$

$$Loss_{img \to txt} = -\frac{1}{N} \sum_{i=1}^{N} \log \frac{\exp(\tau \cdot score(i,i))}{\sum_{j=1}^{N} \exp(\tau \, score(i,j))}$$

$$Loss_{txt \to img} = -\frac{1}{N} \sum_{j=1}^{N} \log \frac{\exp(score(j,j) \cdot \tau)}{\sum_{i=1}^{N} \exp(\tau \, score(i,j))}$$

$$Loss_{total} = \frac{1}{2} Loss_{img \to txt} + \frac{1}{2} Loss_{txt \to img}$$

# NN quantization

weight quant.: FP32 → low_bit (int8, fp4, ...)

goal: memory storage

activation quant: FP32 → low_bit

goal: memory storage + acceleration in computing

1. Post-Training Quantization (PTQ)

$\mathcal{L}(x) \to \min_{x}$, $x$ – full-precision parameters

use $x_q = \text{dequant}(\text{quant}(x))$

2. Quantization-Aware Training (QAT)

$\mathcal{L}(\text{dequant}(\text{quant}(x))) \to \min_{x}$

1. Symmetric quantization

$$\text{quant}(x) = \text{clip}\left(\text{round}\left(\frac{x}{s}\right), q_{min}, q_{max}\right)$$

s – scale parameter

$q_{min} = -127$, $q_{max} = 127$

int8

$$\text{dequant}(q) = s \cdot q$$

2. Asymmetric quantization

$$\text{quant}(x) = \text{clip}\left(\text{round}\left(\frac{x}{s} + z\right), q_{min}, q_{max}\right)$$

$$\text{dequant}(q) = s(q - z)$$

, z – zero-point ← usually rounded to integer

PTQ calibration $\longrightarrow [x_{min}, x_{max}]$

<u>symm. quant.</u> $\alpha = \max(|x_{min}|, |x_{max}|)$

$[-\alpha, \alpha] \longrightarrow [-127, 127]$

$S = \frac{\alpha}{127}$ $\quad$ $\text{dequant}(q) = q \cdot S$

<u>asymm. quant.</u> $S = \frac{x_{max} - x_{min}}{q_{max} - q_{min}}$ $\quad$ $q_{min} = \frac{x_{min}}{S} + Z \Rightarrow$

$\Rightarrow Z = q_{min} - \frac{x_{min}}{S}$

# Calibrations

1. Min-Max observer

$$x_{min} = \min(x_1 \ldots x_N)$$
$$x_{max} = \max(x_1 \ldots x_N)$$

2. Percentile

$$x_{min} = 0.01\text{-perc.}(x_1 \ldots x_N)$$
$$x_{max} = 0.99\text{-perc.}(x_1 \ldots x_N)$$

3. MSE

$$\|x - \text{dequant}(\text{quant}(\text{clip}(x, -\bar{r}, \bar{r})))\|^2 \to \min_{\bar{r}}$$

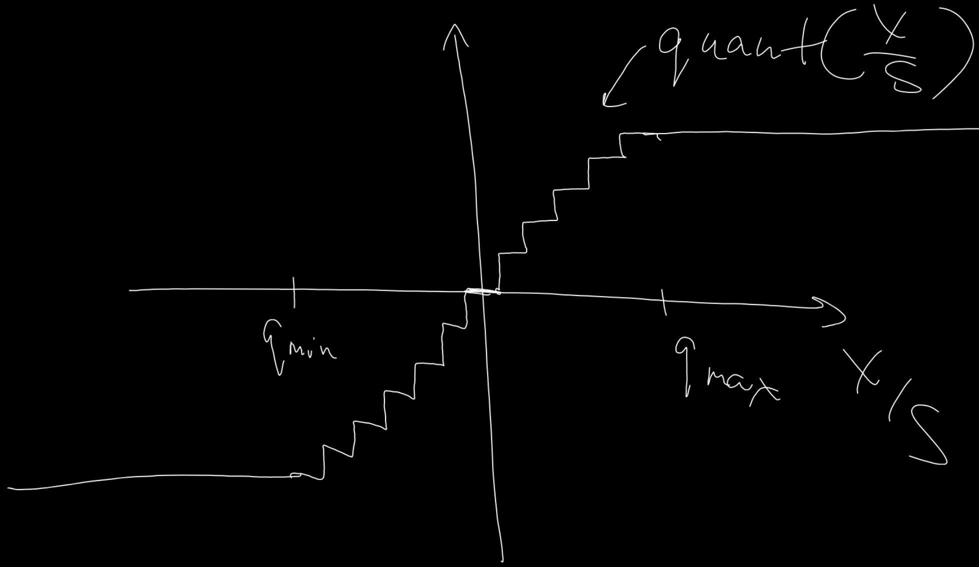
4. $$KL(\text{hist}(x), \text{hist}(\text{dequant}(\text{quant}(x)))) \to \min_{\bar{r}_{min}, \bar{r}_{max}}$$

S / Z : per-tensor, per-channel, per-group

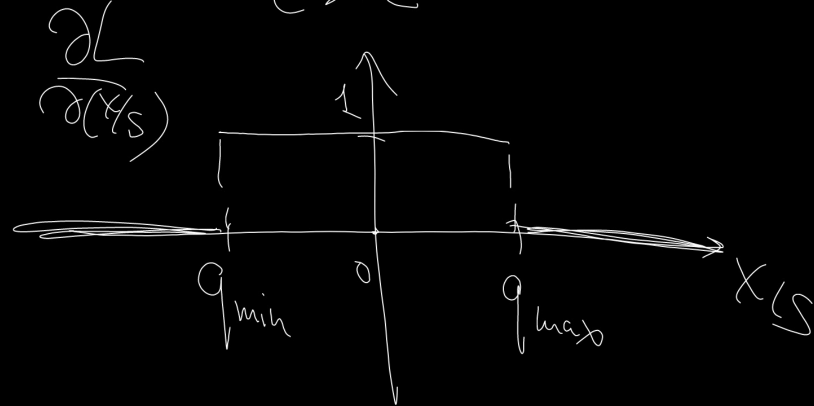
# QAT

$$\mathcal{L}(\text{dequant}(\text{quant}(x))) \to \min_{x,\, s,\, z}$$

$$\text{quant}(x) = \text{clip}\left(\text{round}\left(\frac{x}{s}\right), q_{min}, q_{max}\right)$$




Straight-Trough-Est (STE)



# LoRA and QLoRA

(Low Rank Adaptation)

$$\mathcal{L}(\underbrace{W}_{n\times m} + \underbrace{A}_{n\times r}\cdot\underbrace{B}_{r\times m}) \to \min_{A,B}$$

$r = 8, 16$

QLoRA: $W \xrightarrow{PTQ}$ NF4 (normalized float 4)